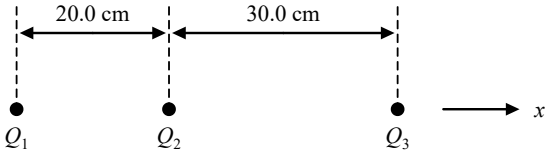
STPM 2022 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $k = 9.0 \times 10^{9}$ N m^2 C^{-2}, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $g = 9.81$ m s^{-2}.

Section A [15 marks]

1. Three charged particles, Q_1, Q_2 and Q_3, of charges −10.0 μC, +5.00 μC and −20.0 μC respectively are arranged in a straight line along the x-axis as shown in the diagram below.



If the distance between Q_1 and Q_2 and the distance between Q_2 and Q_3 are 20.0 cm and 30.0 cm respectively, what is the net force acting on Q_2?

A 1.25 N in the negative-x direction
B 1.25 N in the positive-x direction
C 21.2 N in the negative-x direction
D 21.2 N in the positive-x direction

Answer: A

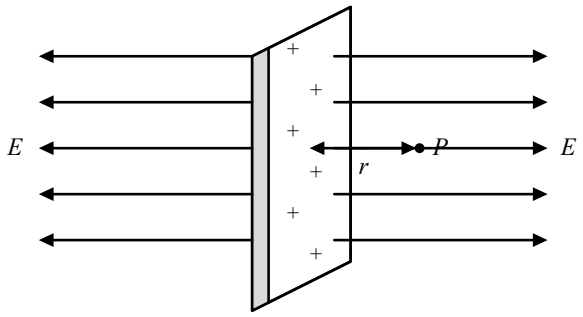
From Q_1 (attraction, $-x$): $\dfrac{(9.0 \times 10^9)(10.0 \times 10^{-6})(5.00 \times 10^{-6})}{(0.200)^2} = 11.25$ N

From Q_3 (attraction, $+x$): $\dfrac{(9.0 \times 10^9)(20.0 \times 10^{-6})(5.00 \times 10^{-6})}{(0.300)^2} = 10.0$ N

Net force = 1.25 N in the negative-x direction

Explanation: Both negative charges attract the positive charge Q_2, in opposite directions.

2. A point, P, is at a distance r from an isolated charged thin metal plate of area A as shown in the diagram below.



If the plate carries charge of $+Q$, what is the electric field strength, E, at P?

A 0 B $\dfrac{Q}{4\pi\varepsilon_o r^2}$ C $\dfrac{Q}{2\varepsilon_o A}$ D $\dfrac{Q}{\varepsilon_o A}$

Answer: C

For a large thin charged plate: $E = \frac{\sigma}{2\varepsilon_o} = \frac{Q}{2\varepsilon_o A}$, independent of r

Explanation: The field near a large plane of charge is uniform.

3. A parallel plate capacitor is connected to a battery. If a dielectric is inserted between the plates of the capacitor while the battery is still connected to the capacitor,

A the potential difference across the capacitor increases
B the energy stored in the capacitor decreases
C the charge stored in the capacitor increases
D the capacitance of the capacitor decreases

Answer: C

With the battery connected the p.d. stays the same; the dielectric increases C, so $Q = CV$ increases.

Explanation: The energy $\frac{1}{2}CV^2$ also increases.

4. A battery of e.m.f. 10 V is connected to a parallel plate capacitor. The area of each plate of the capacitor is 4.0 cm^2. Both plates are separated by air at a distance of 0.25 cm. What is the energy stored in the capacitor when the capacitor is fully charged?

A 7.1×10^{-12} J B 7.1×10^{-11} J C 2.8×10^{-10} J D 2.8×10^{-9} J

Answer: B

$$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(4.0 \times 10^{-4})}{0.25 \times 10^{-2}} = 1.42 \times 10^{-12} \text{ F}$$

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(1.42 \times 10^{-12})(10)^2 = 7.1 \times 10^{-11} \text{ J}$$

Explanation: Convert cm^2 and cm to SI units.

5. Which is true about drift velocity of free electrons in a metal wire after a potential difference is applied across the wire?

A Directly proportional to the number of free electrons per unit volume
B Inversely proportional to the potential difference
C Directly proportional to the cross-sectional area
D Inversely proportional to the resistivity

Answer: D

$J = \sigma E = nev_d \Longrightarrow v_d = \frac{E}{\rho ne}$: inversely proportional to the resistivity

Explanation: For a given p.d. and length, a better conductor gives faster drift.

6. A wire of length 2.0 m has a cross-sectional area of 0.050 cm^2. The resistance of the wire is 0.35 Ω. What is the electrical conductivity of the wire?

A $1.1 \times 10^6\ \Omega^{-1}\ m^{-1}$
B $1.4 \times 10^5\ \Omega^{-1}\ m^{-1}$
C $7.1 \times 10^{-6}\ \Omega^{-1}\ m^{-1}$
D $8.8 \times 10^{-7}\ \Omega^{-1}\ m^{-1}$

Answer: A

$$\sigma = \frac{l}{RA} = \frac{2.0}{(0.35)(0.050 \times 10^{-4})} = 1.1 \times 10^6\ \Omega^{-1}\ m^{-1}$$

Explanation: Conductivity is the reciprocal of resistivity.

7. Which is the basic principle of Kirchhoff's voltage loop rule?

A The conservation of energy
B The conservation of charge
C The conservation of momentum
D The conservation of charge and conservation of energy

Answer: A

The loop (voltage) rule is based on the conservation of energy.

Explanation: The junction rule is based on the conservation of charge.

8. Two resistors of resistance 500 Ω and 250 Ω respectively are connected in series to a battery of 8.0 V with negligible internal resistance. If a voltmeter of resistance of 750 Ω is used to measure the potential difference across the resistor of resistance of 500 Ω, what is the reading of the voltmeter?

A 3.6 V B 4.4 V C 5.3 V D 8.0 V

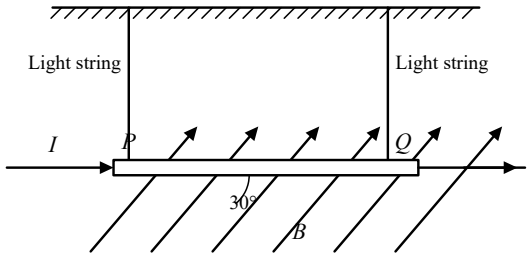
Answer: B

$$500\ \Omega \parallel 750\ \Omega = 300\ \Omega;\ \text{total} = 300 + 250 = 550\ \Omega$$

$$\text{Reading} = \frac{300}{550} \times 8.0 = 4.4\ \text{V}$$

Explanation: The voltmeter draws current, so it reads less than the true 5.3 V.

9. A conductor, *PQ*, of length 50 cm and weight 5.0 N is hung at equilibrium by two light strings. *PQ* is placed in a uniform magnetic field, *B*, of 2.0 T at an angle of 30° as shown in the diagram below.



What is the tension in each string when a constant current, *I*, of 4.0 A flows in *PQ*?

A 0.77 N B 1.5 N C 3.5 N D 4.2 N

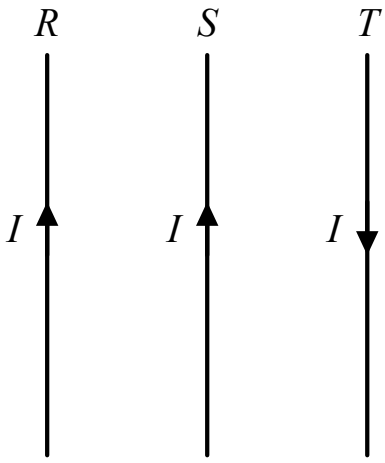
Answer: B

Magnetic force = $BIl \sin 30° = (2.0)(4.0)(0.50)(0.50) = 2.0$ N upward

$2T = 5.0 - 2.0 \Longrightarrow T = 1.5$ N

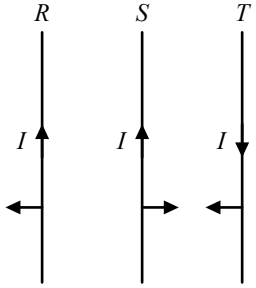
Explanation: Only the field component perpendicular to the wire produces a force.

10. Three parallel long wires, *R*, *S* and *T*, each carries current, *I*, of the same magnitude in the direction as shown in the diagram below.

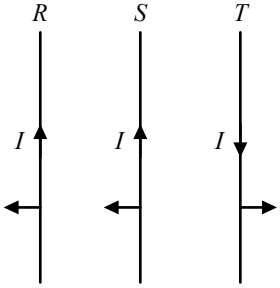


Which diagram shows the correct directions of forces that act on the wires if the distance between *R* and *S* and the distance between *S* and *T* are the same?

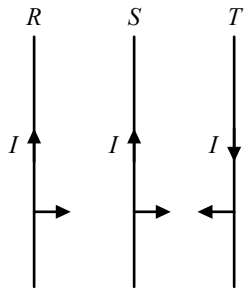
A



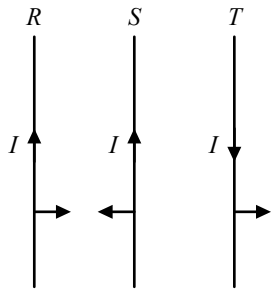
B



C



D



Answer: D

Like currents attract; unlike currents repel. R and S attract; T repels both.

R: pulled right by S, pushed left more weakly by T → right. S: pulled left by R, pushed left by T → left. T: pushed right by both → right (D).

Explanation: The force between wires falls with distance, so the nearer wire dominates.

11. In the Hall effect experiment, when a conductor is placed in a uniform magnetic field, the electric field produced is

A inversely proportional to the current
B directly proportional to the magnetic field
C inversely proportional to the area of the conductor
D directly proportional to the density of charge carriers

Answer: B

$E_H = v_d B$: directly proportional to the magnetic field

Explanation: In equilibrium the electric force on each carrier balances the magnetic force.

12. The current in a 500 turns solenoid of length 8.0 cm increases from zero to 2.5 A in 0.35 s. If the magnitude of the induced e.m.f. is 0.012 V, what is the cross-sectional area of the solenoid?

A 1.7×10^{-3} m^2 B 4.2×10^{-3} m^2 C 2.0×10^{-4} m^2 D 4.3×10^{-4} m^2

Answer: D

$$L = \frac{\varepsilon}{\Delta I/\Delta t} = \frac{0.012}{2.5/0.35} = 1.68 \times 10^{-3}\ \text{H}$$

$$A = \frac{Ll}{\mu_0 N^2} = \frac{(1.68 \times 10^{-3})(0.080)}{(4\pi \times 10^{-7})(500)^2} = 4.3 \times 10^{-4}\ \text{m}^2$$

Explanation: Combine the induced e.m.f. with the inductance of a solenoid.

13. The magnetic flux linkage in an inductor is 5.0 Wb when a constant current of 4.0 A flows in the inductor. What is the energy stored in the inductor?

A 10 J B 20 J C 40 J D 50 J

Answer: A

$$L = \frac{N\Phi}{I} = \frac{5.0}{4.0} = 1.25 \text{ H}; \; U = \frac{1}{2}LI^2 = \frac{1}{2}(1.25)(4.0)^2 = 10 \text{ J}$$

Explanation: Equivalently $U = \frac{1}{2}(N\Phi)I$.

14. A sinusoidal voltage is applied across a resistor. Which statement is true about the relationship between the voltage and the current of the resistor?

A When the voltage is zero, the current is maximum.
B When the voltage is maximum, the current is zero.
C When the voltage increases, the current increases.
D When the voltage decreases, the current increases.

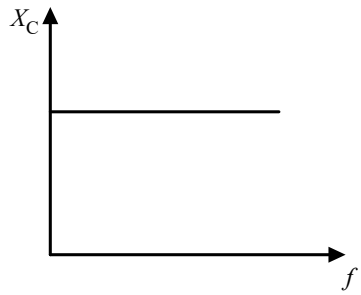
Answer: C

In a resistor the current is in phase with the voltage: as the voltage increases, the current increases.

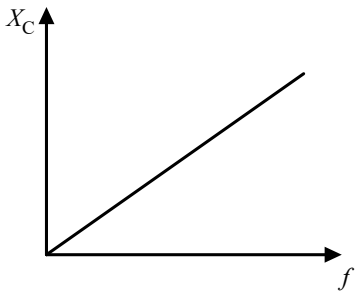
Explanation: $I = V/R$ at every instant.

15. Which graph shows the variation of the reactance of a capacitor, X_C, with the frequency, f, when an alternating voltage is applied across the capacitor?

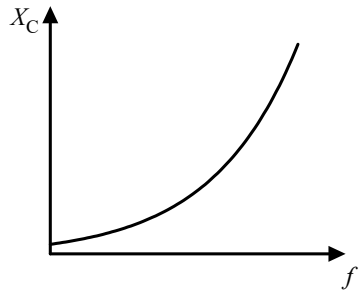
A



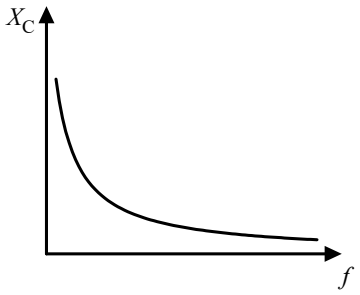
B



C



D



Answer: D

$$X_C = \frac{1}{2\pi fC} \propto \frac{1}{f} \text{ (D)}$$

Explanation: A capacitor passes high frequencies more easily.

Section B [15 marks]

16. An air-filled parallel plate capacitor with the dimension of the plates of 5.0 cm × 5.0 cm has a charge, Q. The electric field between the plates is 3.20×10^5 V m^{-1}. The electric field decreases to 8.89×10^4 V m^{-1} when a dielectric is inserted into the air-filled parallel plate capacitor.

(a) Why does the electric field of the parallel plate capacitor decreases? [1]

Answer

The dielectric is polarised; the induced charges on its surfaces produce a field opposing the original field, so the resultant field decreases — 1

1

Explanation: With Q fixed, E is reduced by the factor ε_r.

(b) Determine Q. [3]

Answer

$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$ — 1

$Q = (3.20 \times 10^5)(8.85 \times 10^{-12})(0.050)^2$ — 1

$Q = 7.08 \times 10^{-9}$ C (7.08 nC) — 1

3

Explanation: Use the field without the dielectric (air).

(c) Determine the dielectric constant. [3]

Answer

$\varepsilon_r = \frac{E_0}{E}$ — 1

$= \frac{3.20 \times 10^5}{8.89 \times 10^4}$ — 1

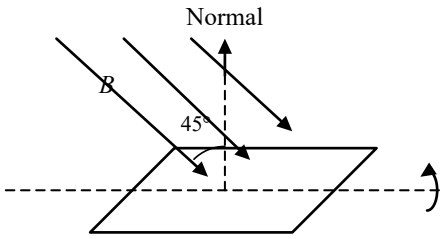
$= 3.6$ — 1

3

Total for question **7**

Explanation: The dielectric constant is the factor by which the field is reduced.

17. A rectangular coil of 5.0 cm × 4.0 cm has 5000 turns and is initially placed with its normal plane set at 45° to the direction of the Earth's magnetic field, B, as shown in the diagram below.



If the total resistance of the coil is 530 Ω and it is rotated about the axis of the plane of the coil by half a turn in 0.100 s, calculate the

(a) change in magnetic flux linkage [4]

Answer

Initial flux linkage $N\Phi_1 = NBA \cos 45°$ — 1

After half a turn the coil faces the opposite way: $N\Phi_2 = -NBA \cos 45°$ — 1

Change $= 2NBA \cos 45° = 2(5000)(5.0 \times 10^{-5})(0.050 \times 0.040) \cos 45°$ — 1

$= 7.1 \times 10^{-4}$ Wb — 1

4

Explanation: Turning the coil through 180° reverses the flux, so the change is twice the initial value.

(b) average e.m.f. induced in the coil [2]

Answer

$\varepsilon = \dfrac{\Delta(N\Phi)}{\Delta t} = \dfrac{7.1 \times 10^{-4}}{0.100}$ — 1

$= 7.1 \times 10^{-3}$ V — 1

2

Explanation: Faraday's law for the average e.m.f.

(c) average current flows through the coil

[The Earth's magnetic field is 5.0×10^{-5} T.] [2]

Answer

$I = \frac{\varepsilon}{R} = \frac{7.07 \times 10^{-3}}{530}$ 1

$= 1.34 \times 10^{-5}$ A 1

2

Total for question **8**

Explanation: Ohm's law for the coil.

Section C [30 marks]

18. (a) Explain the mechanism of conduction of electric current in a metal. [5]

Answer

A metal contains many free (conduction) electrons moving randomly at high speed 1

With no p.d. there is no net drift, so no current 1

A p.d. sets up an electric field in the metal; each electron experiences a force opposite to the field 1

The electrons accelerate between collisions with the lattice ions, then lose energy in collisions (heating the wire) 1

The result is a small average drift velocity opposite to the field, giving a current 1

5

Explanation: $I = nAev_d$.

(b) A potential difference of 2.00 V is applied across an aluminium wire of length 1.00 m and radius 0.050 mm. Calculate the current flows in the wire.

[The electrical conductivity of aluminium is $3.64 \times 10^7\ \Omega^{-1}\ m^{-1}$.] [5]

Answer

$A = \pi(0.050 \times 10^{-3})^2 = 7.85 \times 10^{-9}\ m^2$ 1

$R = \frac{l}{\sigma A}$ 1

$= \frac{1.00}{(3.64 \times 10^7)(7.85 \times 10^{-9})} = 3.50\ \Omega$ 1

$I = \frac{V}{R} = \frac{2.00}{3.50}$ 1

$I = 0.57$ A 1

5

Explanation: Conductivity σ = 1/ρ.

(c) Copper has a molar mass of 6.35×10^{-2} kg and a density of 8900 kg m^{-3}. Assuming that each copper atom contributes one free electron, calculate the drift velocity of the electrons in a copper wire of radius 1.0 mm carrying a current of 0.15 A. [5]

Answer

$n = \dfrac{8900}{6.35 \times 10^{-2}} \times 6.02 \times 10^{23}$ 1

$n = 8.44 \times 10^{28}\ \text{m}^{-3}$ 1

$A = \pi(1.0 \times 10^{-3})^2 = 3.14 \times 10^{-6}\ \text{m}^2$ 1

$v_d = \dfrac{I}{nAe} = \dfrac{0.15}{(8.44 \times 10^{28})(3.14 \times 10^{-6})(1.60 \times 10^{-19})}$ 1

$v_d = 3.5 \times 10^{-6}\ \text{m s}^{-1}$ 1

5

Total for question **15**

Explanation: The drift speed is remarkably small.

19. (a) A long straight wire carries current I. Derive an expression for the magnetic field at a point of distance r from the wire. [3]

Answer

Ampère's law: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$; take a circle of radius r centred on the wire 1

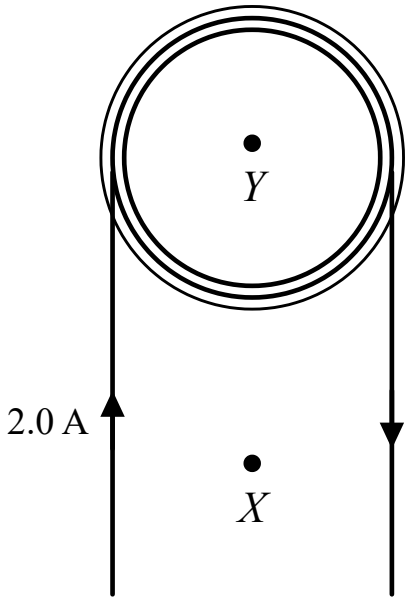
By symmetry B is constant and tangential: $B(2\pi r) = \mu_0 I$ 1

$B = \dfrac{\mu_0 I}{2\pi r}$ 1

3

Explanation: The field lines are concentric circles around the wire.

(b) A long wire which carries current of 2.0 A is looped in the form of circular coil of radius 5.0 cm with 100 turns in a vertical plane as shown in the diagram below.



Determine the magnitude of the magnetic flux density at

(i) the midpoint, X, between the wires [2]

Answer

The two straight leads carry opposite currents 0.10 m apart; at the midpoint their fields add:

$B = 2 \times \dfrac{\mu_0 I}{2\pi(0.050)}$ 1

$B = 2 \times 8.0 \times 10^{-6} = 1.6 \times 10^{-5}$ T 1

2

Explanation: Opposite currents produce fields in the same direction between the wires.

(ii) the centre, Y, of the coil [2]

Answer

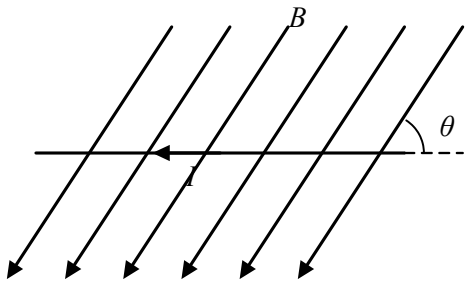
$B = \dfrac{\mu_0 NI}{2r} = \dfrac{(4\pi \times 10^{-7})(100)(2.0)}{2(0.050)}$ 1

$B = 2.51 \times 10^{-3}$ T 1

2

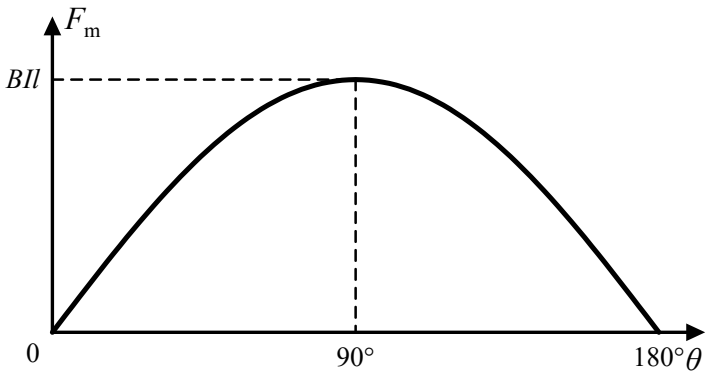
Explanation: Field at the centre of a flat circular coil.

(c) A uniform magnetic field, B, acts on a straight wire that carries a constant current, I, at an angle, θ, as shown in the diagram below.



Sketch a graph of magnetic force, F_m, on the wire against θ, for $\theta = 0°$ to 180°. [2]

Answer



$F = BIl \sin\theta$: zero at 0° and 180°	1
Maximum BIl at 90° (sine shape)	1
	2

Explanation: The force depends on the perpendicular component of the field.

(d) A horizontal copper wire of length 2.0 cm and diameter 1.0 mm carries a current of 2.5 A is placed in the middle of a long solenoid perpendicular to its axis. The solenoid has 4000 turns per metre and carries a steady current, I. Calculate the current in the solenoid if the magnetic force on

the copper wire is equal to its weight.

[The density of copper is 9.0×10^3 kg m^{-3}.] [6]

Answer

Weight of wire = $\rho(\pi d^2/4)lg = (9.0 \times 10^3)\pi(0.50 \times 10^{-3})^2(0.020)(9.81)$ 1

$= 1.39 \times 10^{-3}$ N 1

Magnetic force $BI_w l$ = weight 1

$B = \dfrac{1.39 \times 10^{-3}}{(2.5)(0.020)} = 2.77 \times 10^{-2}$ T 1

Solenoid: $B = \mu_0 nI$ 1

$I = \dfrac{2.77 \times 10^{-2}}{(4\pi \times 10^{-7})(4000)} = 5.5$ A 1

6

Total for question **15**

Explanation: The wire is perpendicular to the solenoid's field, so $F = BIl$.

20. (a) State a difference between a direct current and alternating current. [1]

Answer

Direct current flows in one direction only; alternating current reverses direction periodically 1

1

Explanation: The magnitude of a.c. also varies with time.

(b) State **three** differences between resistance and reactance in an alternating current circuit. [3]

Answer

• Resistance does not depend on frequency; reactance does ($X_L \propto f$, $X_C \propto 1/f$) 1

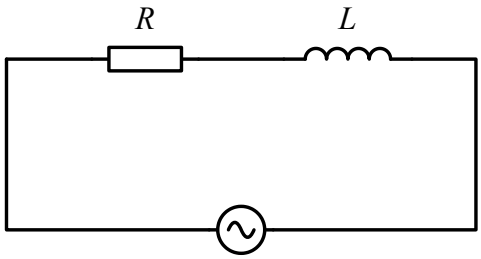
• Resistance dissipates energy as heat; reactance does not (energy is stored and returned) 1

• In a resistance V and I are in phase; in a reactance they are 90° out of phase 1

3

Explanation: Both are measured in ohms.

(c) A circuit consists of an inductor, L, and a resistor, R, which are connected in series to an alternating current supply as shown in the circuit diagram below.



The inductance of L is 35 mH, the resistance of R is 5.0 Ω and the alternating current supply of the root means square voltage is 12 V with a frequency of 50 Hz.

(i) Determine the impedance in the circuit. [4]

Answer

$X_L = 2\pi fL$ 1

$= 2\pi(50)(35 \times 10^{-3}) = 11.0\ \Omega$ 1

$Z = \sqrt{(R^2 + X_L{}^2)} = \sqrt{(5.0^2 + 11.0^2)}$ 1

$Z = 12.1\ \Omega$ 1

4

Explanation: Resistance and reactance combine at right angles.

(ii) Determine the root means square voltage across L. [3]

Answer

$I_{\text{rms}} = \dfrac{12}{12.08} = 0.993$ A 1

$V_L = I_{\text{rms}} X_L = (0.993)(11.0)$ 1

$V_L = 10.9$ V 1

3

Explanation: $V_R = 4.97$ V; the two do not add to 12 V because they are 90° apart.

(iii) Calculate the phase angle of the circuit. [2]

Answer

$\tan \varphi = \dfrac{X_L}{R} = \dfrac{11.0}{5.0}$ 1

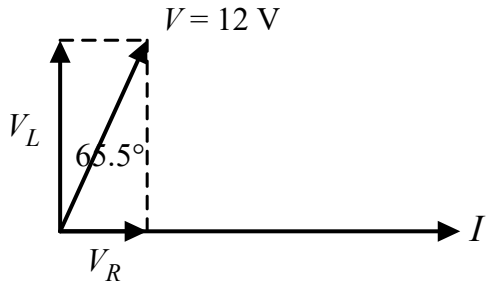
$\varphi = 65.5°$ (voltage leads the current) 1

2

Explanation: In an *RL* circuit the voltage leads the current.

(iv) Sketch the phasor diagram of the circuit. [2]

Answer



V_R in phase with I; V_L leading by 90°	1
Resultant V at 65.5° to I	1
	2
Total for question	**15**

Explanation: Phasors rotate anticlockwise.